

# Midterm 2

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| Current $I$ (A) | B-Field $B$ ( $\mu\text{T}$ ) | Error ( $\mu\text{T}$ ) |
|-----------------|-------------------------------|-------------------------|
| 0.50            | 31.5                          | 0.5                     |
| 1.00            | 62.5                          | 0.5                     |
| 1.50            | 94.5                          | 0.5                     |
| 2.00            | 125.5                         | 0.5                     |
| 2.50            | 157.0                         | 0.5                     |

Table 1: B-field vs. current data.

- What is the direction of the Lorentz force on a negatively charged particle moving in the  $y$ -direction near a positive current in the  $z$ -direction, located at the origin?
- (a) A circular loop of radius  $R = 0.10$  m lies in the  $xy$ -plane and carries a current  $I = 0.5$  A, flowing counterclockwise when viewed from above the plane. What is the magnetic moment vector of the system? (b) The loop is placed in a uniform magnetic field  $\vec{B} = 0.30\hat{i}$  T. Compute the torque vector on the loop. (c) Describe how the loop rotates, and determine its final equilibrium orientation. *Qualitatively, the alignment of molecular and atomic magnetic moments is how magnetization arises in solids.*
- A solenoid of length 20 cm is connected to a DC power supply. A magnetic field probe is placed at the center of the solenoid to measure the B-field. The data are shown in Tab. 1. (a) How many turns are in the solenoid? (b) What B-field is generated with 5.0 A of current? (c) Assuming the model used to solve part (a), what is the reduced  $\chi^2$  value, given the errors?
- (a) An MRI technician moves his hand from a region of very low magnetic field strength into an MRI scanner's 2.00 T field with his fingers pointing in the direction of the field. Find the average emf induced in his wedding ring, given its diameter is 2.20 cm and assuming it takes 0.250 s to move it into the field. (b) What current is induced in the ring if its resistance is  $0.0100\ \Omega$ ? (c) Calculate the average power dissipated in the ring.
- (a) A bicycle generator rotates at 1875 rad/s, producing an 18.0 V peak emf. It has a 1.00 by 3.00 cm rectangular coil in a 0.640 T field. How many turns are in the coil? (b) What is the period of the AC output of the generator?
- Camera flashes charge a capacitor to high voltage by switching the current through an inductor on and off rapidly. In what time must the 0.100 A current through a 2.00 mH inductor be switched on or off to induce a 500 V emf?
- A large research solenoid has an inductance of  $L = 25.0$  H. The current is not switched off instantaneously; instead, it decreases linearly in time according to  $I(t) = I_0(1 - t/\Delta t)$ ,  $0 \leq t \leq \Delta t$ , where  $I_0 = 100$  A and  $\Delta t = 80.0$  ms. (a) Compute the magnitude and sign of the induced emf. (b) Compute the total energy stored in the inductor when the current is at its maximum value. (c) Determine the rate at which energy is dissipated during the shutoff process. For example, compute the instantaneous power  $P(t) = \mathcal{E}(t)I(t)$  and evaluate the average. (d) Compute the reactance of the solenoid at a frequency of 10.0 kHz.
- Your RL circuit has a characteristic time constant of 20.0 ns, and a resistance of 5.00 M $\Omega$ . (a) What is the inductance of the circuit? (b) What resistance would give you a 1.00 ns time constant, perhaps needed for quick response in an oscilloscope? (c) What percentage of the final current  $I_0$  flows 3.0 ns after deactivation?
- Consider Fig. 1. Assume the bottom wire is grounded at 0V. (a) Using Kirchhoff's loop rule on the left side of the circuit, write an equation that describes changes in potential. (b) Repeat exercise (a) for the right loop. (c) Solve parts (a)

and (b) for  $v_{\text{in}}$  and  $v_{\text{out}}$ , respectively, and calculate  $v_{\text{out}}/v_{\text{in}}$  in terms of  $R$  and  $C$ . (d) Is this a low-pass (LP) or high-pass (HP) filter? Argue for LP or HP by *taking the limit* of frequency.

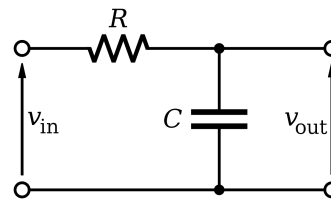


Figure 1: This RC circuit can act as a low-pass filter.

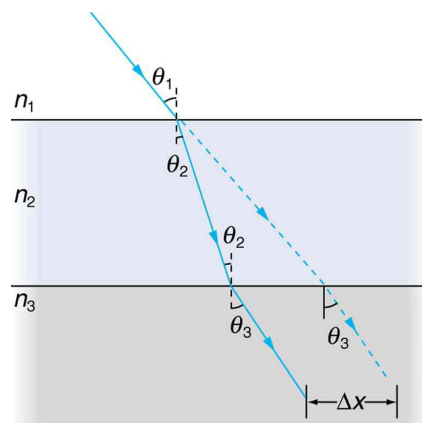


Figure 2: A light ray is in a medium with  $n_1$ , then enters a medium with  $n_2$ , then a medium with  $n_3$ .

appropriate PhET simulation and plot both the simulation data and the data from Tab. 2 on the same graph.

10. (a) What is the resonance frequency,  $f_0$ , of an RLC circuit with  $R = 0.1 \text{ k}\Omega$ ,  $C = 1 \text{ }\mu\text{F}$ , and  $L = 10 \text{ mH}$ ? (b) What is the total impedance of this circuit at  $f = 0.1f_0$  and  $f = 10f_0$ ?
11. For an RLC circuit with that values of  $R$ ,  $L$ , and  $C$  from the previous problem, an AC source voltage characterized by  $V_{\text{rms}} = 120 \text{ V}$ , (a) what is  $I_{\text{rms}}$  at  $3f_0$ ? (b) What is  $P_{\text{rms}}$  at this frequency? *Hint: calculate the power factor, the cosine of the phase angle between resistance and reactance.*
12. (a) What is the B-field generated 1 cm laterally from a capacitor in an RC circuit that charges from 0 to 200 nC in 2  $\mu\text{s}$ ? (b) What *current* is responsible for this B-field, since the capacitor is not passing DC current like a wire?
13. (a) An aircraft receives a pulse reflected from glacial ice below it 10  $\mu\text{s}$  after transmission. What is the displacement between aircraft and ice? (b) What fraction of the power is reflected from the ice, assuming standard indexes of refraction for air and ice at RF wavelengths?
14. (a) A microwave generates 1 kW of power, projected onto a 10 cm-squared area at a distance of 1 m. What is the intensity (power per unit area)? (b) If the intensity drops as the distance squared, what is the intensity 2 meters from the microwave?
15. (a) Show<sup>1</sup> that  $\theta_3 = \theta_1$  in Fig. 2, if  $n_1 = n_3$ . (b) A doctor examines the ear of a child using a 12.0 cm focal length eyepiece 8.0 cm from the ear. What is the magnification?
16. A beam of visible light traveling through air strikes the surface of a transparent material. The angle  $\theta_1$  (measured from the normal) and the refracted angle  $\theta_2$  are measured and shown in Tab. 2. (a) What is the average and standard deviation of the measured index of refraction? (b) Plot  $\sin \theta_2$  versus  $\sin \theta_1$  to show that the data are linearly correlated. Confirm these results with the

| $\theta_1$ (deg) | $\theta_2$ (deg) |
|------------------|------------------|
| 20.0             | 13.2             |
| 30.0             | 19.5             |
| 40.0             | 25.0             |
| 50.0             | 30.0             |
| 60.0             | 35.0             |

Table 2: Incident and refracted angles.

<sup>1</sup>This proof is the reason why the center of our lenses (like those in our glasses) do not change ray directions. If our glasses changed the apparent location of objects, they would be useless.